A New Spatial-Oriented Object Detection Framework for Remote Sensing Images

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***Abstract*— Although the orientation and scale properties of the objects in remote sensing images have been widely considered in the modern deep learning-based object detection methods, the spatial distribution property of objects has rarely been investigated. There is a distinct spatial distribution difference between close-range objects and remote sensing objects: the former may exhibit extensive mutual occlusion and overlap, whereas the latter rarely overlap. A current remote sensing object detection algorithm that ignores the spatial distribution difference may unnecessarily apply the massive anchor-based proposal bounding box generation and nonmaximum suppression (NMS) operations. In this article, considering the unique spatial distribution of remote sensing objects, and also the other spatial properties, we propose a novel, compact, and spatial-oriented object detection framework for remote sensing images. The proposed two-stage convolutional neural network (CNN) framework, which we call the Remote-sensing Spatial Adaptation DETector (RSADet), considers the spatial distribution, scale, and orientation/shape varieties of the objects in remote sensing images. In the first stage, each object instance is inferred on the scale-attention boosted CNN heatmaps to generate candidate bounding boxes, instead of using the anchor-based proposal box generation and NMS. In the second stage, deformable convolutions are introduced to adapt to the geometric variations of different object instances and to avoid the impact of complex and changeable backgrounds. A new bounding box confidence (IoU score) prediction branch is introduced as a convenient constraint for eliminating unreliable boxes and improving performance. Experiments were conducted on a large single-class remote sensing object detection dataset (the Ningbo Pylon dataset) built as part of this study and an open-source extraordinarily large multiclass dataset (the object DetectIon in Optical Remote sensing image (DIOR) dataset). Compared with the advanced detectors from both the computer vision and remote sensing communities, the proposed RSADet achieved state-of-the-art performance on both datasets.**

***Index Terms*— Convolutional neural network (CNN), object detection, object spatial distribution, object-based proposal box inference, remote sensing images.**

I. Introduction

THE detection of objects of interest in remote sensing images has a wide range of applications, including urban environment monitoring, land-use planning, forest fire monitoring, traffic flow management, and military uses [1]–[5]. However, despite the extensive studies conducted, object detection from large-scale and high-resolution remote sensing images remains a great challenge due to the diversity of the objects' scale, orientation, shape, distribution, and illumination condition, and also the complex, confusing, and changing background. As a result, the conventional object detection methods [6], [7] developed based on handcrafted features are concerned with only one or a few properties, such as scale-invariant feature transform (SIFT) [8] and the histogram of oriented gradients (HOG) descriptor [9], may not perform well when nonrigid deformation of object appearance exists. However, significant and rapid progress has recently been made in the development of deep learning [10], [11]. The massive learnable parameters, in the form of multilayer neuron weights in a convolutional neural network (CNN), have made the feature representation ability of the objects in a complex background much stronger. The recent object detection methods in the computer vision field have promoted and inspired the studies of object detection in remote sensing images, the latter have also developed special technologies for the properties of remote sensing objects, including the scale and orientation [12]–[16]. However, we argue in this article that a critical property, i.e., the spatial distribution, which is different in close-range and remote sensing images, has been ignored in these studies, resulting in redundant and unnecessary building blocks in a CNN and relatively low detection performances. In this article, we propose a novel deep learning-based object detection method that is specifically designed for detecting objects in remote sensing images, in which the massive and predefined anchor-based prediction box generation and nonmaximum suppression (NMS) operations are replaced with object-based bounding-box inference (BBI), considering the specific distributions of remote sensing objects (see Fig. 1). Moreover, an attention mechanism for scale invariance, deformable convolutions for orientation and shape diversities, and multihead branches for improving the learning ability of the network under a two-stage detection framework are also presented.

The recent CNN-based object detection methods can be classified into one-stage and two-stage methods. In general, the one-stage methods, such as the well-known you only look once (YOLO) series [17]–[19] and the single shot multibox detector (SSD) [20], usually prioritize the inference speed, while the two-stage methods prioritize the detection precision

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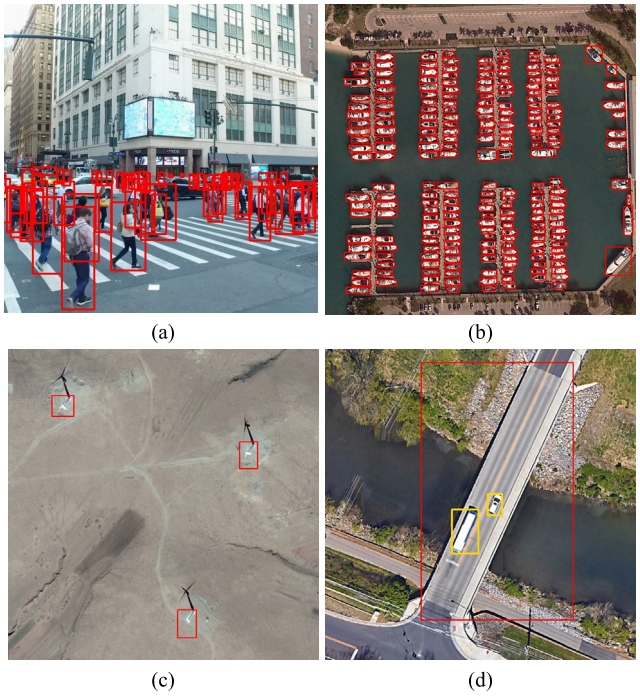


Fig. 1. (a) Crowd in a close-range image. (b) Densely distributed boats. (c) Sparsely distributed windmills. (d) Bridge and vehicles (objects with inclusion relations) in remote sensing images. We argue that, even in the cases of (b) and (d), anchor-based massive prediction box generation and NMS operations are unnecessary.

and recall. The one-stage methods predict the categories and bounding boxes of the objects of interest directly in a single network. The one-stage detectors [19], [20] still need to produce dense candidate boxes on the high-level feature maps; however, only a few boxes contain objects, in most cases, and the background samples dominate the training process and lead to model degradation. To address this problem, RetinaNet [21] introduces focal loss to solve the extreme imbalance between the foreground and background classes, while CornerNet [22] and CenterNet [23] generate fewer bounding boxes by estimating the key points of objects, rather than defining a set of default anchor boxes. The former predicts the top-left and bottom-right corners of each potential object, and the latter predicts the location of an object's center point and regresses its size.

The method proposed in this article lays more stress on accuracy and belongs to the two-stage methods. The mainstream two-stage methods generate massive anchor-based proposal boxes via the so-called region proposal network (RPN) in the first stage, and then regress the refined bounding boxes and identify the categories of objects in the second stage. The R-CNN series are typical two-stage methods. R-CNN [24] consists of region proposal generation (by selective search [25]), feature vector extraction (by AlexNet [26]), candidate region classification (by support vector machine or SVM), and bounding box refinement (by linear regression). Fast R-CNN [27] introduces a region of interest (ROI) pooling layer to produce a fixed-length feature vector. Faster R-CNN [28] introduces the anchor-based RPN, which predefines a set of default anchor boxes of different scales and ratios for each pixel on the feature maps as proposal boxes for the objects, to replace the selective search strategy. The RPN has become the footstone of the current two-stage object detection methods. The feature pyramid network (FPN) [29] constructs a feature pyramid structure in Faster R-CNN for better detection of multiscale objects. Region-based fully convolutional network (R-FCN) [30] designs position-sensitive score maps to handle the dilemma between translation-invariance in image classification and translation-variance in object detection. Mask R-CNN [31] adds a parallel mask prediction branch to Faster R-CNN. Developed from Mask R-CNN and the FPN, path aggregation network (PANet) [32] shortens the information propagation path between the lower layers and the topmost layers.

The deep learning-based object detection methods have also been extensively developed in the remote sensing community [33]–[37]. However, the objects in remote sensing images have different characteristics than close-range objects. In particular, the multiscale and multiorientation properties have attracted extensive attention [38], [39].

The vast scale diversity is derived from the different object categories, the different object instances of the same category, and the different spatial resolutions of the images. To handle this multiscale problem, multilevel feature fusion methods have been applied in the CNN-based remote sensing object detection methods. For example, Qian *et al.* [40] proposed a multilevel feature fusion module; based on YOLOv2, Liu *et al.* [41] concatenated the multilayer feature maps at the route layer; Huang *et al.* [42] proposed a cross-scale feature fusion pyramid network (CF2PN) to address the large variations in object sizes; Fu *et al.* [14] combined the feature maps of the top layers and shallow layers via a top-down pathway and a bottom-up pathway. Other works in the remote sensing field that have tackled the multiscale problem include the studies by Ding *et al.* [43], Lin *et al.* [44], Zhang *et al.* [45], and Huyan *et al.* [46].

For the problem of the arbitrary orientation of remote sensing objects, two approaches have typically been applied. One approach utilizes oriented anchor boxes and the other adopts special convolution operations. Li *et al.* [47] added extra multiangle anchors into the RPN module; Fu *et al.* [12] built a rotation-aware object detector with multiple default angles and oriented ROI pooling; Liu *et al.* [48] designed a similar rotated region-based CNN; Xu *et al.* [49] utilized deformable convolutions [50] to model the geometric variations of remote sensing targets. Specifically, deformable convolution adds 2-D offsets into the common convolution grid to ensure that the feature sampling procedure is conducted in unfixed locations, which enhances the ability of the detector to recognize the orientation and shape variations of objects.

Differing from these works focusing on the challenges from the scale, orientation, or shape varieties of single objects, we are concerned with the specific spatial distribution of object instances in remote sensing images. This is a critical property, but has not been investigated in the remote sensing field in the age of deep learning. The distinct spatial distribution difference between close-range and remote sensing objects impels us to reconsider whether the most widely used anchor-based detector, is really needed or even harmful.

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There are densely distributed objects in remote sensing images [as shown in Fig. 1(b)], but they seldom overlap. The mainstream two-stage object detection methods for remote sensing images [16], [42], [51], [52] all utilize per-pixel anchor-based RPNs and the subsequent NMS operation. However, many of the dense anchor boxes can be invalid. This is also the case for the anchor-based one-stage methods [41], [42]. We argue that dense and predefined anchor-based proposal box generation and NMS are unnecessary and even inappropriate when detecting objects in remote sensing images. By removing these operations, the network can be greatly simplified, and the efficiency and performance of the object detection methods can be improved. To the best of our knowledge, the inadaptability between the spatial distribution property of objects in remote sensing images and the predefined anchor-based proposal box generation has not been investigated. Anchor-based proposal box generation and NMS have been used in all of the recent object detection methods for remote sensing images [14], [16], [42], [43], [52]. Instead, we regard the object-based strategy, which first recognizes the existing objects instead of predefining per-pixel anchor boxes in an image (or a feature map) and then generates candidate bounding boxes of these objects, as being much more suitable for remote sensing object detection.

In this article, a novel two-stage object detection framework based on an object-based BBI strategy that meets the spatial distribution property of remote sensing objects is proposed. Furthermore, the proposed framework is also a compact and neat synthesis of the different approaches that have been separately developed for handling the scale diversity, the orientation and shape arbitrariness, and the spatial distribution properties of remote sensing objects. As such, the proposed two-stage object detection framework represents a new baseline structure for object detection in remote sensing images. Specifically, an object-based BBI module is developed from CenterNet to predict the objects' centers and sizes; a scale attention module is designed to enhance the multiscale representation ability; and a series of deformable convolutions are introduced in the second stage to increase the detector's robustness to the arbitrary orientation and variation of the background.

There are also other practical improvements for further boosting the object detection performance, i.e., an intersection over union (IoU) score prediction branch is newly added to eliminate unreliable candidate proposals; the combination of smooth L_1 loss [28] and complete IoU (CIoU) loss [53] results in more accurate regressed bounding boxes; and a top k loss-based hard sample mining strategy is proposed to discover and handle hard samples, to improve the training efficiency. Finally, postprocessing consisting of overlapped cropping and a global Soft-NMS operation [54] is adopted to rerecognize individual and complete objects in the large remote sensing images, instead of small image tiles, which is a necessary step but is one that has been ignored in recent remote sensing detection research.

Moreover, we created a new pylon dataset covering a geographically continuous area of 331 km^2, which is a beneficial supplement to the existing remote sensing object detection datasets such as [55] from two aspects. First, pylon is not included in these datasets; second, pylon is such a typical object that is sparsely distributed, but these existing datasets provide object instances in the form of small cropped and discrete patches, ignoring the real situation that some types of objects are sparsely distributed in vast and complex backgrounds.

The main contributions of this article are summarized as follows.

1) For the first time, the spatial distribution property of objects in remote sensing images is investigated in a deep learning framework, and an object-based BBI module based on object center estimation is correspondingly introduced. The anchor-based RPN and NMS operations used in the recent remote sensing object detection methods are reasonably rejected.
2) An integrated and spatial-oriented framework is proposed for handling the scale, orientation and shape, and spatial distribution properties of remote sensing objects. Furthermore, the proposed framework possesses a lightweight, neat, and compact structure, and represents a new baseline structure for object detection in remote sensing.
3) Practical improvements include a newly introduced IoU score prediction branch and a top k loss-based hard sample mining strategy for achieving a better detection effect, and a postprocessing to recover individual objects in the large-size remote sensing images from cropped image tiles.
4) We edited and released a large-scale electricity pylon detection dataset (http://gpcv.whu.edu.cn/data/), the first open-source aerial pylon dataset called the Ningbo Pylon dataset. It is a very challenging dataset, and even challenging for a human to label the secluded and scattered pylons.
5) For both the Ningbo Pylon dataset and the extraordinarily large multiclass object DetectIon in Optical Remote sensing image (DIOR) dataset [55], the proposed method comprehensively and significantly outperformed the state-of-the-art object detection methods in both the computer vision and remote sensing communities. This further demonstrates the critical importance of handling the spatial distribution property of remote sensing objects in a modern deep learning framework.

II. Methods

A. Proposed Framework

The goal of this work is not only to emphasize the spatial distribution property of the objects in remote sensing images, which has been ignored in the previous object detection studies, but also to establish a new integrated, comprehensive, and compact baseline structure that synchronizes the cutting-edge technologies and ideas for boosting generic network performance and handling the scale and orientation properties of remote sensing objects. To achieve this, the proposed framework, which we call the Remote-sensing Spatial Adaptation DETector (RSADet), is designed in a modular manner, and each module has an explicit purpose. As shown in Fig. 2, an object-based BBI module is applied to predict

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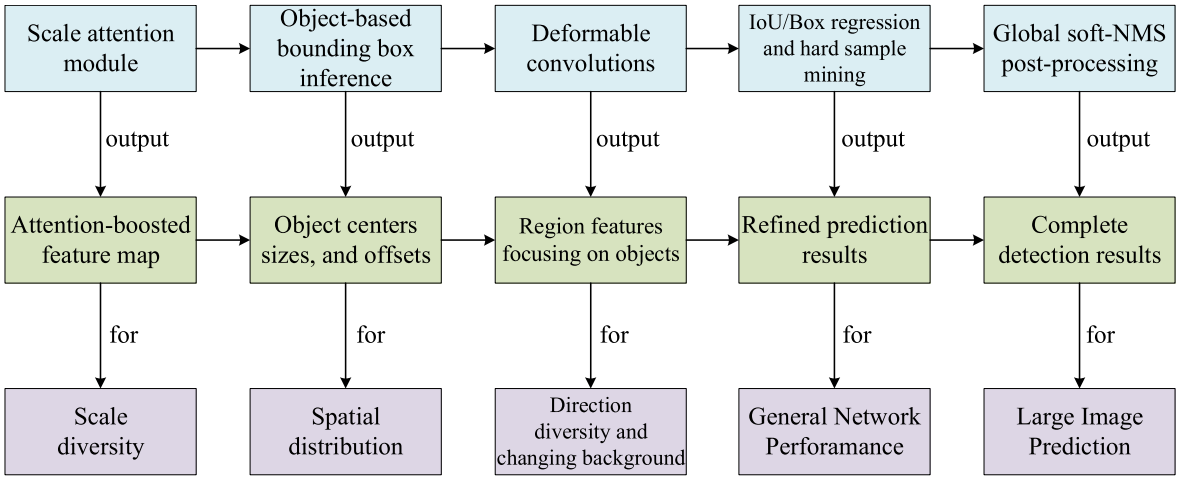


Fig. 2. Main modules in the proposed framework and their purposes.

the centers and sizes of the candidate objects, without the use of NMS, for the unique spatial distribution property of remote sensing objects. Embedded in the frontend of the BBI module, a lightweight scale attention module is used to handle the spatial scale diversity. For the arbitrary directions and variable shapes of objects, a secondary encoder consisting of a series of deformable convolutional layers is applied, which also has resistance to changeable and complex backgrounds. For general network performance boosting, we introduce two operations. An IoU score prediction branch is newly added to compensate the common regression loss for the four variables that record the center, width, and height of the bounding boxes, remove invalid candidate boxes, and obtain refined bounding boxes. A hard sample training strategy is also applied throughout the training phase, as confusable objects and objects blending into the background are especially common in remote sensing images, and require more attention than the easily distinguishable objects. The final module is designed to fit outside of the trainable network, for the practical purpose of detecting complete objects in a large-capacity remote sensing image restored from the image tiles cropped for the deep learning model prediction.

Fig. 2 depicts the main modules in the proposed framework, and the structure of the executable and compact two-stage object detection network is presented in Fig. 3. In the first stage, the network takes an optical remote sensing image as the input, extracts scale attention boosted multilayer convolutional neuron features, estimates the object centers and sizes of all the classes, and generates candidate bounding boxes. The candidate bounding boxes and their corresponding feature maps are then fed into the ROIAlign layer to produce fixed-size local pooling features. In the second stage, the secondary encoder consists of three simple deformable convolutional layers that extract higher-level features of the objects from the background through irregular sampling operations from the local pooling features. There are also two shared fully connected layers and two independent layers each for the IoU score branch and box regression branch that map the extracted features to the expected outputs. The IoU score prediction branch estimates the IoU score between the candidate box and the corresponding ground-truth box, which reflects the coincidence degree of the predicted boxes. The box regression branch predicts the four offsets between the candidate box and the ground-truth box with the newly designed combined loss function. Both branches adjust the candidate box in the training loop and finally output the refined bounding box coordinates.

B. Multiscale Feature Extraction

In this work, differing from most of the previous works that use a feature pyramid to handle the variety of the objects' spatial scales, the representation of the multiscale property is realized by a scale attention module that is embedded in the multilayer neuron feature extractor. The attention module operates on the $4\times$ downsampled feature map and generates an attention weight map to emphasize the features of objects at different scales.

First, we chose the lightweight deep layer aggregation (DLA) network [56] (DLA-34 in our study) with hierarchical skip connections as the backbone feature extractor. DLA iteratively and hierarchically fuses the information across layers and merges the features with different resolutions and scales. Given an $\in R^{H\times W\times 3}$ (where H, W, and 3 separately represent the height, width, and channel number of the image), the DLA backbone outputs a feature map $F \in R^{h\times w\times n}$ with a downsampling factor of 4, where $h = H/4$, $w = W/4$, and n is the channel number. We refer the reader to [56] for more details of the DLA network. For the feature map, we introduce a scale attention module to highlight the multiscale representations of objects.

The scale attention module is composed of a multiscale feature extraction submodule, a spatial-channel pooling submodule, and an attention map generation submodule, as presented in Fig. 4. The multiscale feature extraction submodule consists of three convolutional layers. To reduce the computational cost, the first layer is a single 3×3 convolution with a channel reduction rate of four to compress the channel number of the input feature, which is followed by a rectified linear unit (ReLU) activation operation. The second layer consists of four parallel convolutions equipped

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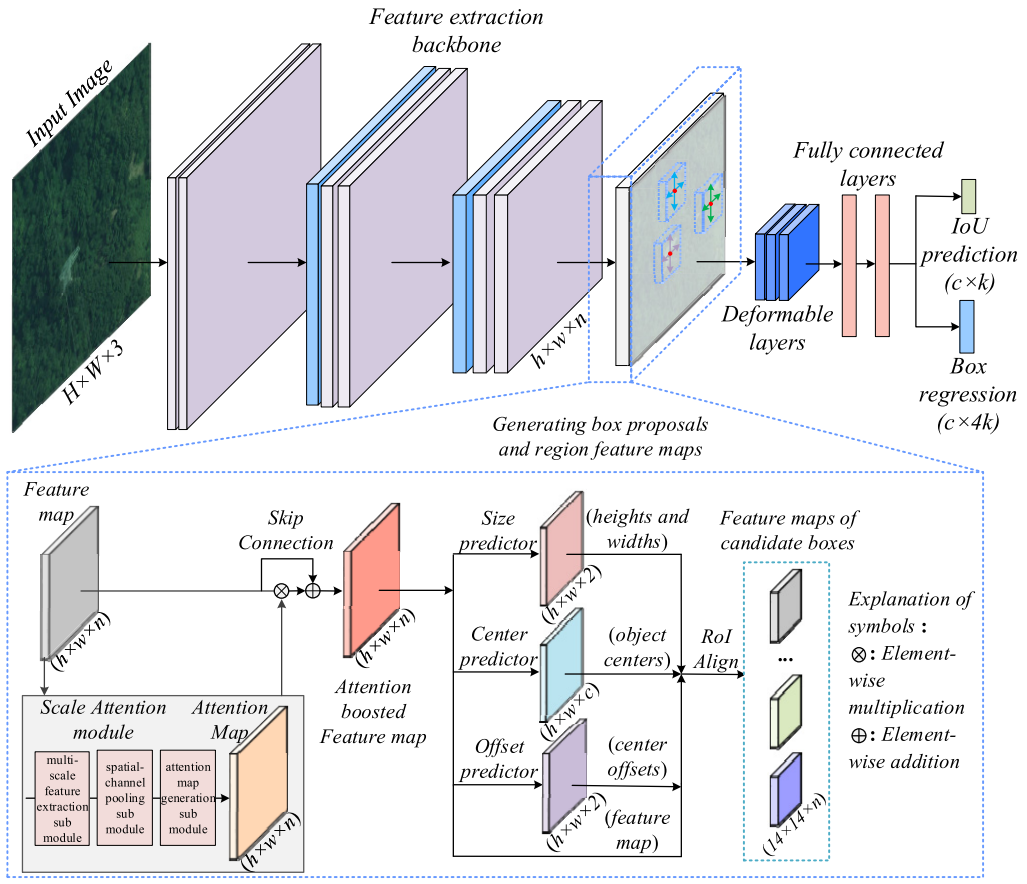


Fig. 3. Proposed two-stage object detection network for remote sensing images. c is the category number and k is a preset number, in that only the top k candidate boxes are processed.

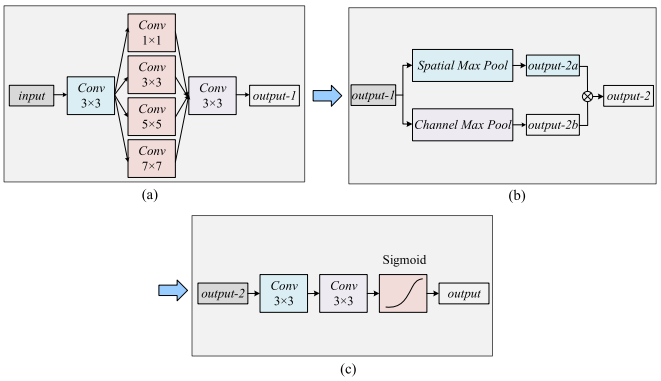


Fig. 4. Workflow of the proposed scale attention module. (a) Multiscale feature extraction sub module. (b) Spatial-channel pooling sub module. (c) Attention map generation sub module.

with different kernel sizes and respective ReLU activations. We adopt 1×1, 3×3, 5×5, and 7×7 convolution kernels to explicitly extract multiscale features from the view angles of different receptive fields. The outputs of the four convolutions are concatenated together and fed into the third layer, which is a single 3×3 convolution with the channel number of the input of the attention module and a ReLU.

The spatial-channel pooling submodule [Fig. 4(b)] contains global maximum pooling operations along both the spatial and channel dimensions. An average pooling operation is not adopted as the peaks of the subsequent feature map (positions with the maximum response value among the eight neighbors) should be reserved and extracted as object centers. The pooled features along the spatial ($F_s \in R^{1\times1\times n}$) and channel ($F_c \in R^{h\times w\times1}$) dimensions are, respectively, expanded to the same shape of $h \times w \times n$ in a duplication manner. Element-wise multiplication between them is then conducted to generate the spatial-channel descriptor, i.e., F_{sc} ($\in R^{h\times w\times n}$) $= F_s \otimes F_c$. The attention map generation submodule [Fig. 4(c)] takes F_{sc} as the input and yields the attention weight map. It consists of two 3×3 convolutions, with a ReLU for the nonlinear representation ability of features and a sigmoid activation function for normalizing the weight map, respectively. Compared with the commonly used feature pyramid strategy with all sorts of skip connections and fusions, the scale attention module is more concise and convenient.

Finally, the attention weight map is first multiplied by and then element-wise added to the input of the attention module, to obtain the final attention-boosted feature map, as shown in Fig. 3.

C. Object-Based BBI Module

The object-based BBI module directly detects latent bounding boxes of objects from the attention-boosted feature map. In object detection, an object can be described by the two

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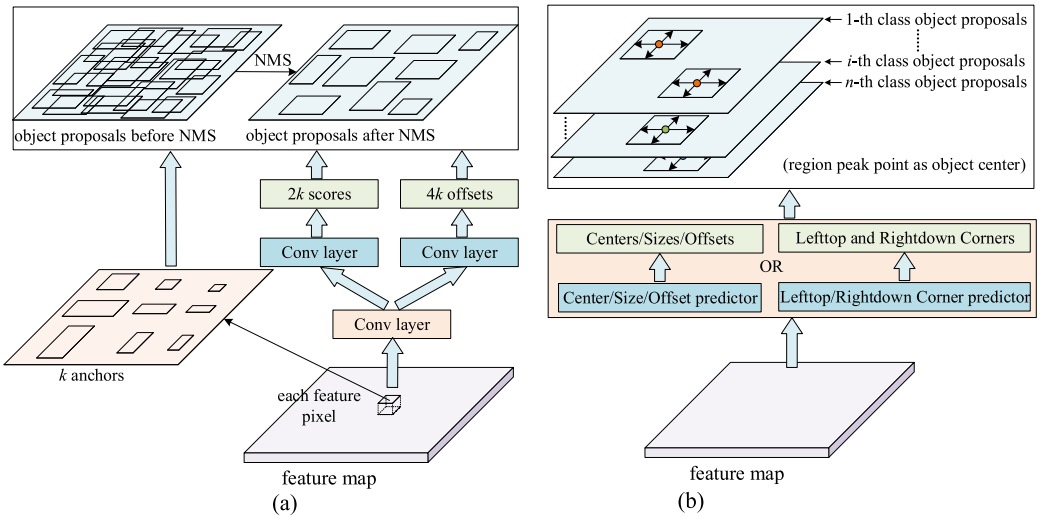


Fig. 5. Difference between the anchor-based and the object-based methods. The former predefines k fixed-shape dense proposal boxes (anchors) for each pixel and picks out those boxes that more likely contain an object (without category information) using NMS; the latter directly predicts the bounding boxes of category-specified objects. (a) Anchor-based method. (b) Object-based method.

diagonal corners or the center point with width and height. We chose the latter form, as in Zhou *et al.* [23], and regress the object center, size (width and height), and offset with parallel and independent predictors. The offset records the rounding error. All of the predictors consist of a 3×3 convolution and a 1×1 convolution, and they output the corresponding prediction heatmaps. The center-point heatmap with a shape of $h \times w \times c$ predicts the probability of position (x_i, y_i) being the center of an object of each category, where c is the number of categories. The offset heatmap with a shape of $h \times w \times 2$ regresses the offsets $(\Delta x_i, \Delta y_i)$ of the object center coordinates. The size heatmap with a shape of $h \times w \times 2$ regresses the height h_i and width w_i of the object centered at the position of (x_i, y_i). Finally, the top k peaks in the center-point heatmap of all the categories are selected as the candidate object centers; for example, 128 was used in the DIOR dataset during the training stage, and 300 during the test stage in the experiments. The coordinates of a candidate bounding box (i.e., the top-left corner and bottom-right corner) can be easily calculated according to the object center and its width and height. At the end of the module, each candidate bounding box is mapped onto the final attention-boosted feature map to crop the corresponding region, which is then resampled to a fixed-size region feature map ($14 \times 14 \times 64$ in our case) via the well-known ROIAlign operation [11] as the box-wise input of the second stage.

Differing from the object-based BBI, the widely used anchor-based RPN [28] is based on predefined dense proposal boxes. There is also another important difference between the two strategies. In the anchor-based RPN, the proposal network only judges the probability of a box containing an object, and it requires an extra branch for predicting the object's category. In contrast, the center-point heatmap with c (the number of all categories) channels predicts and records the probability of a box containing a category-specified object. Thus, objects with inclusion relations [see Fig. 1(d)] can be independently detected without interference. The two differences are also shown in Fig. 5.

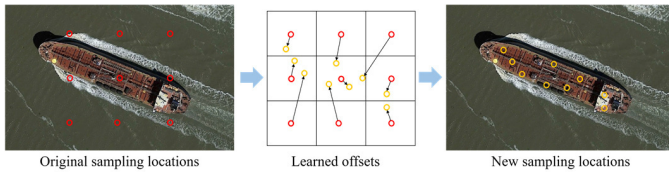


Fig. 6. Illustration of the deformable convolution sampling procedure.

D. Refinement Stage

The second stage (the refinement stage) is designed to enhance the object detection performance, and considers both the geometric localization accuracy and semantic recognition precision in the cheapest but most effective way. Each feature map of the inferred bounding boxes is fed into the proposed detection module, which consists of three deformable convolutions to extract higher-level semantic features and two fully connected layers to transform the semantic features to the expected outputs (i.e., the bounding boxes of the objects). The deformable convolution adds extra 2-D offset learning parameters to adjust and aggregate the convolutional kernels onto the foreground object regions, as shown in Fig. 6. Hence, the deformable convolution can better handle the geometric variations of objects, such as the shape and orientation, and can resist the complex and changeable background.

The predicted outputs are supervised and iteratively corrected by the ground-truth bounding boxes with losses from two branches. In addition to the common box regression branch, we also introduce a new IoU score prediction branch. This branch estimates the category-specific IoU scores between the predicted boxes and the ground-truth boxes of all c categories, i.e., the output of the IoU prediction branch for each candidate box is a c-dimension vector. The reason we

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introduced this constraint is that we discovered that it can effectively remove the invalid candidate boxes; if a candidate box does not overlap with any ground-truth bounding boxes, its IoU score should be zero.

The smooth L_1 loss is adopted to optimize the parameters of the IoU score prediction branch. The smooth L_1 loss is defined as follows:

$$\text{Loss}_{\text{IoU}} = \text{Smooth}_{L_1}(x) = \begin{cases} 0.5\ x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \quad (1)$$

where x denotes the difference between the predicted IoU score and the real IoU value between the predicted box and the ground-truth box.

The commonly used box regression branch tunes the candidate box coordinates. Following the work of R-CNN [24], the mapping from a predicted box to the target box is parameterized to four translation functions. The regression targets can be calculated as follows:

$$t_x = (G_x - P_x)/P_w, \quad t_y = \left(G_y - P_y\right)/P_h \quad (2)$$

$$t_w = \log(G_w/P_w), \quad t_h = \log(G_h/P_h) \quad (3)$$

where t_x, t_y, t_w, and t_h are the regression parameters, and P_x, P_y, P_w, P_h and G_x, G_y, G_w, G_h denote the center-point coordinates, width, and height of the predicted box and the ground-truth box, respectively. Only those predicted boxes with an IoU score (calculated with the corresponding ground-truth box) greater than 50% are used in the regression procedure.

The existing object detection approaches usually adopt l_n-norm loss to optimize the four parameters (i.e., t_x, t_y, t_w, and t_h) independently. However, the four variables are actually correlated. Regressing these variables separately makes some bounds of the predicted boxes accurate, but others not. In contrast, the IoU-based loss functions, such as the latest CIoU loss [53], use the IoU metric as a factor to measure the distance between the predicted and target box, and have been proven to be beneficial for box regression. The formula for CIoU loss is defined as follows:

$$L_{\text{CIoU}} = 1 - IoU + \frac{\rho^2\left(b^{pt}, b^{gt}\right)}{c^2} + av \quad (4)$$

where $\rho^2(b^{pt}, b^{gt})$ is the Euclidean distance between the center points of the predicted and the ground-truth box, c is the diagonal line length of the smallest enclosing box of the two boxes, and av is used to keep the consistency of the height-to-width ratio. In addition

$$a = \frac{v}{1 - \text{IoU} + v}, \quad v = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w^{pt}}{h^{pt}}\right)^2. \quad (5)$$

However, CIoU loss has its weakness, in that the height-to-width ratio discrepancy between the predicted and the target boxes cannot fully reflect the real size difference, and CIoU loss may optimize the predicted box to the target one with a similar aspect ratio but incorrect size. In fact, the smooth L_1 loss [28] can supervise the size problem. Therefore, in this work, the combination of the smooth L_1 loss and CIoU loss is introduced to optimize the box regressor as follows:

$$\text{Loss}_{\text{box}} = \text{Smooth}L_1 + L_{\text{CIoU}} \quad (6)$$

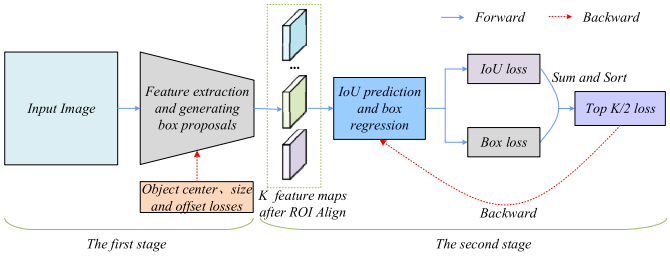


Fig. 7. Forward pass and backward pass operations when training the proposed two-stage object detection framework. The gradient signals of the backward pass are truncated at the ROIAlign operation.

Note that the smooth L_1 loss used here compares the predicted box parameters t_x', t_y', t_w', t_h' with the ground truth, which is different from the smooth L_1 loss for the IoU score branch.

E. Hard Sample Mining

In remote sensing images, there is likely to be a small number of confusable objects and objects that are indistinguishable from the background (i.e., hard samples); however, there is always a large number of easy samples. A training procedure that does not consider this imbalance may lead to reduced effectiveness and slower convergence. The loss values of the easy examples comprise the majority and dominate the gradient updating, whereas the hard samples with high losses are marginalized. In this work, a hard sample mining strategy is introduced to improve the learning efficiency. In the work of [57], the online hard example mining strategy (OHEM) was designed to make the training procedure of the two-stage detectors more effective and efficient. In OHEM, candidate boxes are scored by their loss values, and NMS is applied to select the samples with the highest losses. The highest loss samples constitute a minibatch, which is used to train the classification and regression subnetworks of the second stage. The other thousands of candidates with low losses and high overlap with the selected high-loss samples are removed. However, in our work, we just generate dozens or hundreds of object candidates without NMS and overlap. It is, however, necessary to modify OHEM to make it suitable for the proposed remote sensing object detection framework.

We simplified the OHEM strategy to a top k loss-based hard sample mining strategy, and removed the read-only branch without backpropagation [57] and the loss-based NMS operation. The IoU prediction loss and the box regression loss of each candidate proposal are summed. All the candidates are then sorted by the summed losses, and the top k candidates (hard samples) are selected for backward propagation at both the IoU prediction branch and the box regression branch. The training procedure is detailed in Fig. 7. For the N candidate proposals generated from the first stage, we select the half ($k = N/2$) of the candidates with the highest losses to update the model parameters. The two stages are trained simultaneously. The gradient signals of the backward pass are then truncated at the ROIAlign operation.

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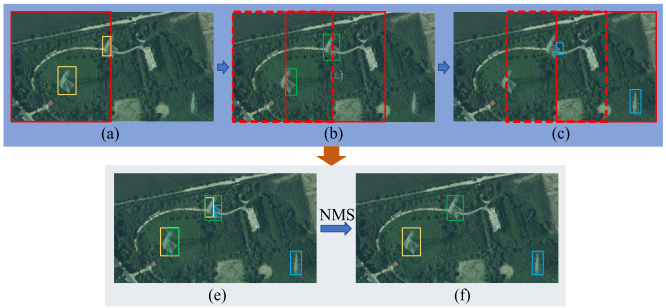


Fig. 8. Postprocessing using Soft-NMS for eliminating the redundant boxes predicted from the adjacent image tiles with 50% overlap. (a)–(c) Object detection results from different tiles with different colors. (d) All of the detected boxes. (e) Results after Soft-NMS and only the best bounding boxes remain.

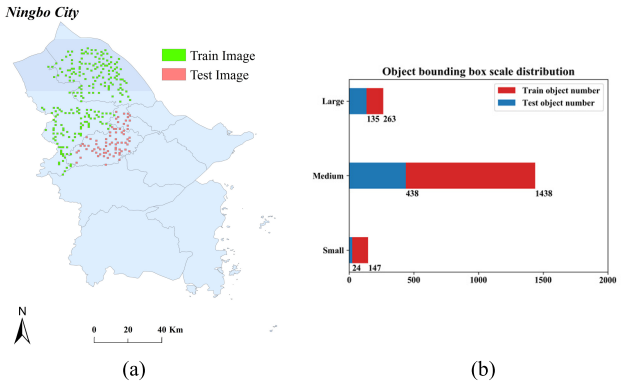


Fig. 9. (a) Geographic locations of the images in the Ningbo Pylon dataset and (b) scale distribution of the object bounding boxes according to the MS COCO partition criteria.

F. Preprocessing and Postprocessing

Aerial or satellite remote sensing images possess a far larger size than natural images, and before being input into a CNN-based detector, they should be cropped into tiles, to adapt to the GPU capacity. As a result, some individual objects are inevitably cut into parts in different tiles. Parts of an object not only cause human semantic ambiguity but also confuse the CNN model. To address this and prepare the training dataset (the Ningbo Pylon dataset), we cropped the remote sensing images with a suitable side length and stride (1024 and 512 pixels, respectively) to ensure that most of the complete object instances appear within at least one cropped tile. For the other small parts of objects (less than half of the current object's area), we simply masked them with black pixels to ensure that the detector would concentrate on the complete objects during the training.

For the prediction/test procedure, overlapped cropping with a suitable stride (e.g., 50% overlap) is important to obtain complete object bounding boxes. When predicting in new images, the parts of objects in different tiles may be recognized several times as individual objects. We, therefore, designed a postprocessing procedure to handle the redundant prediction boxes. All the boxes generated from different tiles are mapped back onto the original image, and each box is scored with its predicted IoU value, which reflects how well the box covers the object. The Soft-NMS algorithm [54] is then employed to preserve the best bounding boxes iteratively in the large-scale remote sensing image. The low IoU scores are further decreased after the Soft-NMS operation, and we remove the corresponding boxes by a preset threshold. The designed postprocessing procedure is summarized in Fig. 8.

III. Experiments and Results

The comparison between the proposed RSADet method and other state-of-the-art CNN-based methods from both the computer vision and remote sensing fields was conducted on two remote sensing object detection datasets. The first dataset was the single-category Ningbo Pylon dataset, which was built and released by ourselves. The other dataset was the large public DIOR dataset with multiple categories [55]. Details of the two datasets are provided in Section III-A, the evaluation metrics are introduced in Section III-B, the results of the different methods are provided in Section III-C, and a detailed analysis, including ablation experiments, is provided in Sections III-D–III-F.

A. Datasets

1) Ningbo Pylon Dataset: We created the Ningbo Pylon dataset for developing supervised object detection methods for remote sensing images, to make up for the current lack of open-source high-resolution pylon datasets. The Ningbo Pylon dataset is a large-scale and challenging dataset consisting of 304 6000 × 6000-pixel aerial images with a ground sampling distance of 0.2 m. There is a total of 2445 pylon instances with manually edited bounding box annotations. The aerial images, which were captured in May 2019, cover parts of the cities of Cixi (a county-level city) and Yuyao (a county-level city), and also Haishu District and Jiangbei District of the city of Ningbo (a prefecture level city) in Zhejiang province, China, as shown in Fig. 9(a), with a total area of around 331 km^2. In this dataset, the 228 images of the cities of Cixi and Yuyao are used for the training set, and the 76 images of Haishu District and Jiangbei District are used for the test set. When preparing the training set, the tiles which had no pylons in them were abandoned, and a total of 5727 1024 × 1024-pixel tiles were finally obtained. The optimal training hyperparameters were obtained by using 30% of the samples as the validation set, and then all of the training samples were used for training the compared models. The test images are not cropped, so that, in the testing stage, they can be just cropped into 1024 × 1024-pixel tiles with a 25% overlap online and fed into the detector. The detection results of the tiles can then be seamlessly reunited with the global Soft-NMS algorithm to output the final complete prediction map. According to the Microsoft Common Objects in Context (MS COCO) partition criteria [58], most of the objects in this dataset are of medium size. The distributions of the large (area $\geq 96^2$ pixels), medium (32^2 pixels $\leq$ area $< 96^2$ pixels), and small (area $< 32^2$ pixels) objects are shown in Fig. 9(b), and some examples are displayed in Fig. 10.

Pylons are typically sparsely distributed over a large area, and manually labeling them is time-consuming, laborious, and prone to omission. We took three steps to carefully prepare this dataset in an interactive and semiautomatic manner.

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Fig. 10. Examples of pylon instances with annotated bounding boxes. The first row is samples of large-size objects, the second row is samples of medium-size objects, and the last row is samples of small objects.

First, the historical pylon positions (geographic points), which were available in the historical records, were marked in the georeferenced images to manually delineate the bounding boxes of the known instances. The outdated records (i.e., where no pylons exist in the images) were deleted, and some omitted pylons were also labeled by the first-round checking of all the images. Second, the images with box annotations were then used to train the proposed two-stage detector. The predicted bounding boxes were generated by setting a low precision threshold, to ensure a high recall rate. Third, each box was carefully checked in the second round, the false detection boxes were removed, and the bounding box annotations were adjusted when necessary. The proposed detector helped us to find 169 pylons missing from human eyes.

2) DIOR Dataset: The DIOR dataset [55] is an extraordinarily large scale remote sensing object detection benchmark dataset, which consists of 23 463 800 × 800-pixel images and 192 472 axis-aligned object bounding box annotations, covering 20 object categories of interest. The dataset is divided into a training-validation set (11 725 images) and a test set (11 738 images). The ground resolutions of the images in the dataset range from 0.5 to 30 m, and all of the images were collected from Google Earth. The densely distributed objects, the large interclass and intraclass size variations, the different spatial resolutions and imaging conditions, as well as other factors, make this dataset very challenging. Examples of the 20 classes are shown in Fig. 11.

B. Evaluation Metrics and Settings

The average precision (AP) and the mean AP (mAP) are taken as the evaluation metrics, which are widely adopted in object detection studies for both natural and remote sensing images. The AP is counted by calculating the area under the precision-recall curve (recall varies from 0% to 100%). The precision and recall are defined as follows:

$$\text{Precison} = \frac{\|\text{TP}\|}{\|\text{TP}\| + \|\text{FP}\|} \tag{7}$$

$$\text{Recall} = \frac{\|\text{TP}\|}{\|\text{TP}\| + \|\text{FN}\|}. \tag{8}$$

A predicted bounding box with an IoU score exceeding the predefined threshold is recorded as a true positive instance (TP), or otherwise a false positive (FP). False negative (FN) denotes that the ground-truth bounding box has not been detected. The $||\cdot||$ symbol means counting the number of all the TP (or FP, FN) instances. Given different IoU thresholds, the AP value changes. In general, a higher IoU threshold leads to a lower AP value. The higher AP under a given IoU threshold denotes a better performance.

For the Ningbo Pylon dataset, we varied the IoU thresholds from 0.50 to 0.95 at an interval of 0.05 to count the corresponding AP values and then the mAP, as commonly used. The mAP, AP_{50} (IoU threshold = 50%), and AP_{75} (IoU threshold = 75%) values are reported to comprehensively evaluate the performance of the different algorithms. The DIOR dataset is more complex and the objects are hard to accurately localize, due to the large interclass diversity and intraclass similarity. Therefore, we used an IoU threshold of 0.5 to determine the TP instances and then calculated the AP_{50} for each category and the mAP_{50} for all the categories, which is consistent with the approach used in the DIOR dataset article [55].

In the following parts, all the experiments with the Ningbo Pylon dataset were executed on a Windows PC equipped with an NVIDIA GeForce GTX 1080 Ti GPU and an Intel Core i5-8400 CPU. The experiments with the DIOR dataset were

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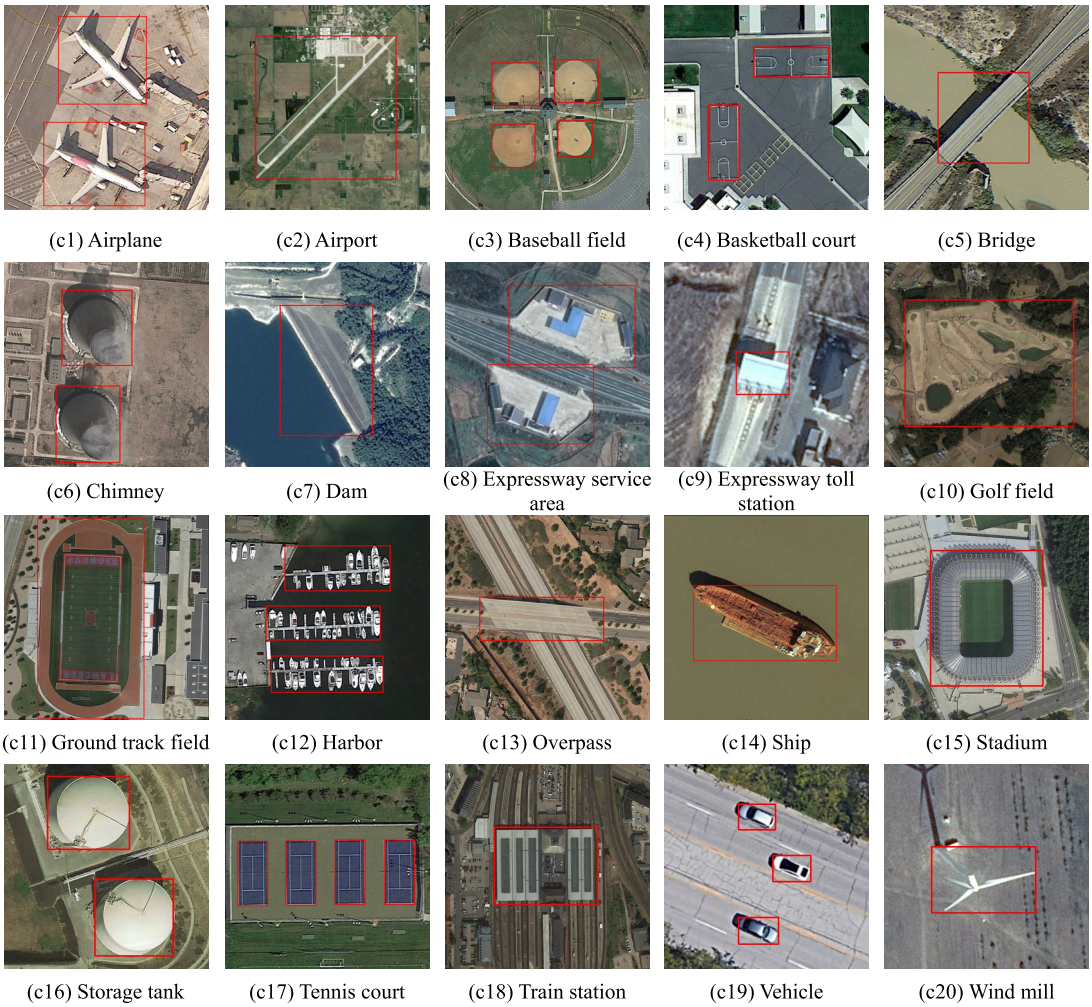


Fig. 11. Examples of each category and their bounding boxes (red rectangles) in the DIOR dataset.

carried out on a more powerful Windows PC equipped with an NVIDIA TITAN RTX GPU and an Intel Core i9-9900K CPU, to save time. The commonly used data augmentation strategies, including random flipping, resizing, shifting, and rotating, were applied on the training samples for all the methods. For the Ningbo Pylon dataset, the large-size aerial images are provided for the performance evaluation. The global Soft-NMS postprocessing procedure was conducted for all the methods.

C. Comparison With the State-of-the-Art Methods

1) DIOR Dataset: We comprehensively compared the proposed RSADet method with the recent state-of-the-art object detection methods from both the computer vision and remote sensing fields on the DIOR dataset with the default training-validation set and test set partition. The Faster R-CNN [28], PANet [32], YOLOv4 [59], CenterNet [23], and RetinaNet [21] methods are from the computer vision community, where Faster R-CNN and PANet are two-stage methods and the rest are one-stage methods.

The local contextual feature fusion network (LCFFN) [47], context aware (CA)-CNN [60], polymorphic module and group attention module boosted (PG)-YOLO [61], contextual bidirectional enhancement (CBD-E) [15], and CF2PN [42] methods are from the remote sensing community. PG-YOLO and CF2PN are one-stage methods and the other methods are all two-stage methods. All of the methods are dense anchor-based methods. PG-YOLO introduces a polymorphic module and group attention module to learn the features of objects with different sizes and shapes, based on YOLOv3. CF2PN utilizes a cross-scale fusion module to extract multiscale semantic features. CBD-E uses a gated function to filter out unexpected background information and spatial-group-based attention to emphasize the foreground, based on the FPN [29]. Both CA-CNN and LCFFN are contextual feature fusion-based methods designed to improve the detection performance of Faster R-CNN. Clearly, these methods are only concerned with parts of the spatial properties of the objects in remote sensing images.

The AP (AP_{50}) for each category (the full name of each category can be found in Fig. 11) and the mAP for all the categories (mAP_{50}) of the different methods are listed in Table I. The proposed RSADet method achieves a mAP of 72.2% and outperforms the best-performing computer vision method (CenterNet) by 2.8% and the best-performing remote sensing method (CBD-E) by 4.4%. The proposed RSADet method also shows the best performance at the category level. In nine categories, the proposed RSADet method performs

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TABLE I
AP OF EACH CLASS AND THE MAP OF ALL THE CLASSES FOR THE DIFFERENT METHODS ON THE DIOR DATASET

Network	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15	c16	c17	c18	c19	c20	mAP_{50}
Faster R-CNN	53.6	49.3	78.8	66.2	28.0	70.9	62.3	69.0	55.2	68.0	56.9	50.2	50.1	27.7	**73.0**	39.8	75.2	38.6	23.6	45.4	54.1
PG-YOLO	71.5	74.8	75.3	87.8	34.6	72.0	42.0	66.2	68.5	61.9	62.6	49.8	49.1	88.8	66.6	73.2	87.0	44.3	49.1	82.8	65.4
YOLOv4	58.5	61.5	65.2	86.4	34.4	72.1	46.2	70.9	47.8	63.5	73.3	55.6	48.9	89.5	41.7	71.3	85.7	40.9	54.7	85.0	62.7
RetinaNet	53.3	77.0	69.3	85.0	44.0	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
PANet	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.0
CA-CNN	54.3	76.8	64.4	80.5	46.2	72.1	59.5	76.1	64.5	77.3	79.0	44.2	57.2	67.5	70.2	60.1	83.2	64.1	41.0	82.9	66.1
LCFFN	54.6	77.1	75.8	87.4	40.4	**76.7**	**63.8**	71.5	62.7	76.4	78.1	43.8	57.0	74.5	68.5	64.1	84.0	56.3	42.9	**86.8**	67.1
CF2PN	**78.3**	78.3	76.5	88.4	37.0	71.0	59.9	71.2	51.2	75.6	77.1	56.8	58.7	76.1	70.6	55.5	**88.8**	50.8	36.9	86.4	67.3
CBD-E	54.2	77.0	71.5	87.1	44.6	75.4	63.5	76.2	65.3	**79.3**	**79.5**	47.5	59.3	69.1	69.7	64.3	84.5	59.4	44.7	83.1	67.8
CenterNet	74.7	80.1	**79.2**	89.4	**47.2**	75.0	54.6	74.5	68.4	79.1	79.2	41.4	**61.0**	66.3	65.5	65.7	88.7	56.3	**55.3**	85.5	69.4
RSADet	73.6	**86.0**	72.6	**89.6**	43.6	75.3	62.3	**79.5**	**68.7**	78.6	79.1	**57.9**	59.2	**90.0**	55.8	**77.0**	87.8	**65.3**	**55.3**	86.5	**72.2**

the best, whereas the second-best method (CenterNet) only leads in four categories. In particular, the proposed RSADet method achieves the best results for the most densely distributed objects, including ships (c14), vehicles (c19), and storage tanks (c16), which clearly demonstrates that the object-based strategy is more suitable for detecting objects in remote sensing images, and that the anchor-based region proposal strategy and NMS operation are unnecessary, even in such densely distributed cases.

This finding is further supported by the observations made on the different performances of the anchor-based methods and the object-based methods. The two object-based methods (the proposed RSADet method and CenterNet) perform clearly better than the anchor-based methods, even though the former methods utilize a light feature extractor (DLA-34), whereas many of the latter use a much heavier extractor. For example, RetinaNet, PANet, and CBD-E use ResNet-101, and CF2PN uses the backbone based on M2Det [62]. In fact, the anchor-based methods can outperform the object-based methods when being applied on a close-range dataset instead of a remote sensing dataset. This strong evidence demonstrating the object detection difference between close-range and remote sensing images is further discussed in Section III-D in detail.

Another observation is that the methods especially designed for the detection of objects in remote sensing images (i.e., CA-CNN, LCFFN, CF2PN, and CBD-E) perform almost the same, indicating that they have exhausted the common spatial properties, such as the scale, context, and orientation, and have reached a ceiling. These methods do outperform some recent computer vision methods (RetinaNet and PANet), but not by much. In contrast, the proposed RSADet method holds a safe lead over all the other methods.

Examples of the detection results of the proposed RSADet method are shown in Fig. 12, with both the predicted bounding boxes and their categories. Objects with different scales, orientations, and spatial distributions can be recognized and localized reliably. In the first line of Fig. 12, the densely distributed ships (c14), storage tanks (c16), and vehicles (c19) can be finely located. The overpass and the cars on it can also be separately detected, without interference (the last image of the last row).

2) Ningbo Pylon Dataset: In the experiment with the Ningbo Pylon dataset, the two-stage methods of Faster R-CNN [28], Faster R-CNN embedded with OHEM [57], and PANet [32], and the one-stage methods of RetinaNet [21], YOLOv4 [59], and CenterNet [23], were applied. The different backbones used by the different methods are listed in Table II. Except for the proposed RSADet and CenterNet, all the other methods use an anchor-based bounding box localization mechanism. For a fair comparison, we only considered methods providing source codes.

Table II lists the mAP, AP_{50}, and AP_{75} of the experimental results of the different methods. Faster R-CNN performs the worst among the seven CNN-based detectors. The OHEM strategy improves the detection precision and recall of Faster R-CNN by 1.9% in terms of mAP. RetinaNet performs the best among the anchor-based methods; however, its superiority is not that obvious when compared to YOLOv4 and PANet. The object-based methods (the proposed RSADet and CenterNet) comprehensively outperform the anchor-based methods, demonstrating once again the advantage of the object-based bounding box localization strategy for remote sensing images. The proposed RSADet outperforms the second-best CenterNet by 3.1%, 1.0%, and 5.5% in items of mAP, AP_{50}, and AP_{75}, respectively. Note that the RSADet shows a significant advantage over the other methods at the high IoU threshold AP_{75} by outperforming CenterNet by 5.5% and the best anchor-based method (RetinaNet) by 12.2%, indicating that the proposed RSADet can predict much more precise bounding boxes than the other methods.

The efficiencies of the different methods to process a 6000×6000-pixel aerial image are listed in Table II. Faster R-CNN, CenterNet, and the proposed RSADet are almost the same in terms of speed, but are slightly slower than YOLOv4. PANet and RetinaNet with heavy backbones and dense anchor-based multiscale prediction are less efficient. However, considering the large size of the remote sensing images, all the detectors are efficient. The parameter capacity of our method is 50 Mb and on par with that of the mainstream methods.

Examples of the object bounding boxes generated by the RSADet are shown in Fig. 13, where the predicted IoU score from the IoU score prediction branch and the real IoU between the detected box and the ground-truth box are annotated. The predicted boxes (orange rectangles) overlap well with the ground-truth bounding boxes (green rectangles). The estimated IoU scores approach the real IoU, so the IoU

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Fig. 12. Examples of the detection results of the proposed RSADet method with the DIOR dataset. The predicted bounding boxes and their categories are annotated in the original images. The negative detection boxes are labeled as green boxes, the missing detection boxes are red boxes, all of them are not attached with category annotations.

TABLE II

COMPARISON OF THE DIFFERENT METHODS ON THE NINGBO PYLON DATASET

Network	*Backbone*	*mAP*(%)	AP_{50}(%)	AP_{75}(%)	Params(M)	*Cost time*(*s*)
Faster R-CNN	ResNet-101	48.5	89.2	49.8	47.27	9.46
Faster R-CNN+OHEM	ResNet-101	50.4	89.9	52.3	47.27	9.46
PANet	ResNetXt-101	52.7	90.2	57.0	291.94	16.74
RetinaNet	ResNet-101	55.8	89.3	62.1	55.43	15.82
YOLOv4	CSPDarkNet-53	54.8	92.3	61.9	60.04	7.56
CenterNet	DLA-34	58.9	93.0	68.8	20.17	9.22
RSADet	DLA-34+SAM	**62.0**	**94.0**	**74.3**	50.37	10.67

score prediction branch therefore forms a strong constraint to recognize the imperceptible and dim pylon instances, such as those in the last row of Fig. 13, and to eliminate false objects.

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Fig. 13. Examples of bounding boxes from the proposed RSADet. The ground-truth boxes (green boxes), the predicted boxes (orange boxes), as well as the real IoU values (the text in the green background) and the predicted IoU values (the text in the orange background) are annotated.

TABLE III

DETECTION RESULTS OF DIFFERENT DETECTORS ON THE MS COCO TEST-DEV DATASET

Network	*Backbone*	*mAP*(%)	AP_{50}(%)	AP_{75}(%)
PANet	ResNet-50	42.5	62.3	46.4
PANet	ResNetXt-101	47.4	67.2	51.8
RetinaNet	ResNet-101	39.1	59.1	42.3
RetinaNet	ResNetXt-101	40.8	61.1	44.1
CenterNet	DLA-34	39.2	57.1	42.8
CenterNet	Hourglass-104	42.1	61.1	45.9

TABLE IV

DETECTION RESULTS OBTAINED ON THE NINGBO PYLON DATASET WITH DIFFERENT BACKBONE NETWORKS

Network	*Backbone*	*mAP*	AP_{50}(%)	AP_{75}(%)
Proposed	DLA-34	62.0	94.0	74.3
Proposed	Hourglass-104	63.7 (+1.7)	93.1	77.1

D. Comparison Between Remote Sensing and Close-Range Images

The spatial distributions of the objects in close-range images and remote sensing images are significantly different.

TABLE V

COMPARISON OF ANCHOR-BASED AND OBJECT-BASED STRATEGIES, DEFORMABLE CONVOLUTIONS AND REGULAR CONVOLUTIONS IN THE PROPOSED NETWORK ON THE PYLON DATASET

Variation	*Candidate box generation*	*mAP*	AP_{50}(%)	AP_{75}(%)
First encoder	Anchor-based	53.6	89.1	61.5
	Object-based	**61.6**	**93.8**	**74.1**
Secondary encoder	Regular convolutions	60.9	93.5	72.0
	Deformable convolutions	**61.6**	**93.8**	**74.1**

We therefore need a specific candidate bounding box generation strategy, which is the main argument of this article. In this section, we compare the performances of the different methods on the remote sensing datasets used in the experiments described previously and a close-range dataset, to further demonstrate this difference. Fortunately, we did not need to run these methods on the MS COCO dataset as many studies have reported their results on it [21], [23], [32]. The MS COCO dataset containing 118k training images, 5k validation images, and 20k test images (test-dev) is a famous and challenging

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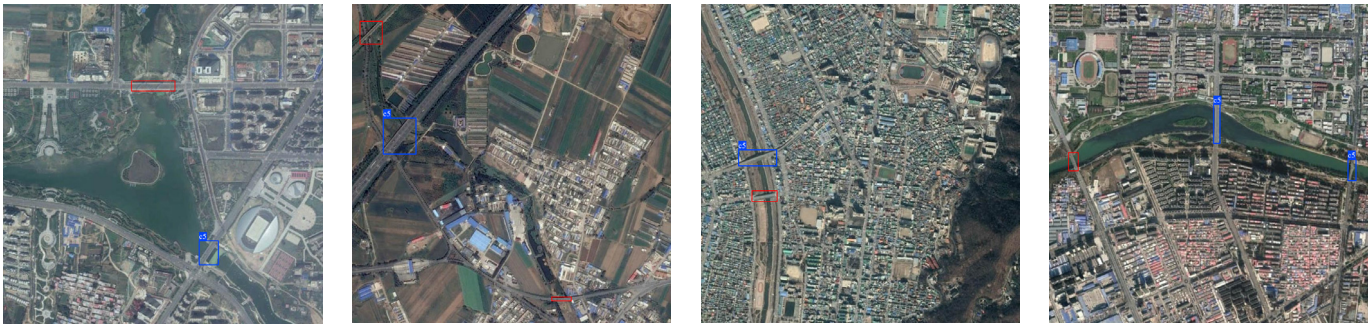


Fig. 14. Examples of bridge detection results of the proposed RSADet in the DIOR dataset. The positive detections are labeled by blue boxes, the missing ones by red boxes.

TABLE VI
ABLATION STUDY RESULTS OF THE PROPOSED DETECTION FRAMEWORK ON THE NINGBO PYLON DATASET

Network	*Box branch Smooth L_1*	*Box branch CIoU*	*IoU branch Smooth L_1*	*Hard sample mining*	*mAP*(%)	*AP_{50}*(%)	*AP_{75}*(%)
One-stage baseline	-	-	-	-	60.0	93.9	70.4
Baseline+parts	√	-	-	-	60.5	93.3	71.1
	-	√	-	-	60.7	93.3	71.3
	√	√	-	-	61.0	93.8	73.2
	√	√	√	-	61.6	93.8	74.1
	√	√	√	√	**62.0**	**94.0**	**74.3**

benchmark dataset for object detection and segmentation in natural images.

As shown in Table III, PANet, which is a representative state-of-the-art anchor-based two-stage method, outperforms the one-stage methods of CenterNet (object-based) and RetinaNet (anchor-based) by a large margin. Specifically, PANet with ResNetXt-101 outperforms CenterNet with DLA-34 by 8.2% in mAP on the close-range MS COCO dataset. In contrast, with the same backbones, CenterNet outperforms PANet by 3.4% and 6.2% in mAP for the remote sensing DIOR (Table I) and Ningbo Pylon (Table II) datasets. This clearly demonstrates that the object-based strategy is far more suitable than the anchor-based strategy for detecting objects from remote sensing images. However, before our work, this had not been recognized by the remote sensing community, and all of the recent remote sensing methods (as listed in Table I) applied the anchor-based strategy and NMS, which are in fact only suitable for close-range object detection.

E. Heavier Backbone

In this article, our aim is to propose a novel, neat, and compact baseline framework with only the necessary elements for object detection in remote sensing images. In fact, we could easily magnify our competitive advantage against the current state-of-the-art computer vision or remote sensing methods by, for example, using a popular heavier backbone network, as most of the remote sensing object detection methods do [15], [21], [41].

To validate this assumption, the Hourglass backbone [63], which combines bottom-up and top-down processing and an intermediate supervision mechanism, was applied in the proposed detector. As shown in Table IV, compared with the lightweight backbone of DLA-34, the results obtained when using Hourglass-104 achieve a 1.7% mAP improvement with the Ningbo Pylon dataset. That is to say, the proposed method now outperforms the state-of-the-art methods by at least 4.8% in mAP, by comparing Tables II and IV.

F. Effectiveness of the Individual Parts

To further demonstrate the effectiveness of the anchor-based method in remote sensing object detection, we replaced the object-based BBI module in the first encoder of our method with the anchor-based one [28]. As shown in Table V, the performance of anchor-based method is significantly lower than the object-based one, demonstrating again the object-based method is much more suitable for the spatial distribution property of remote sensing objects. The comparison of deformable and ordinary convolutions in the secondary encoder is also shown in Table V. Compared to the ordinary convolutions, the deformable convolutions enhance the detection performance by 0.7%, 0.3%, and 2.1% in terms of the mAP, AP_{50}, and AP_{75}, respectively.

The contributions of the rest individual parts are shown in Table VI. The backbone is the one-stage object-based detector embedded with the scale attention module, where the introduction of the scale attention module handling the multiscale property of remote sensing objects results in 1.1% mAP improvement. The bounding box regression branch enhances the mAP by 0.5%, 0.7%, and 1.0% with the smooth L_1 loss, the CIoU loss, and their combination, respectively, demonstrating the effectiveness of the combined loss function. The further introduction of the IoU score prediction branch provides an additional improvement in mAP of 0.6%. The hard sample mining strategy further increases the mAP by 0.4%.

G. Limitations

Object detection from remote sensing images is still a challenging problem as there may be confusable objects and

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complex backgrounds. For example, the proposed method obtained the lowest score in the category of bridge according to Table I, although it beat most of the other methods. Bridges are tightly connected with roads, and it is hard to distinguish bridges from roads in many situations from bird-eye remote sensing images. Fig. 14 shows such examples. We may not fully depend on the current data-driven deep learning-based methods; how to embed human knowledge, for example, the contextual relationship between bridges and roads, into a deep learning model, may be a new direction of remote sensing object detection.

IV. Conclusion

In this article, a novel object-based two-stage object detection baseline framework (RSADet) has been proposed to detect the objects of interest in remote sensing images. We have shown in this article, for the first time, that the object-based BBI method is much more effective than the anchor-based methods in object detection from remote sensing images, considering the distinct spatial distribution difference between close-range objects and remote sensing objects. Remote sensing objects seldom overlap; in this situation an object-level strategy is more suitable than a pixel-level strategy, the latter is applicable to objects with large overlaps. When compared with the results of the state-of-the-art methods from both the remote sensing and computer vision fields obtained on two large-scale remote sensing datasets, the advantages of the object-based strategy and the whole framework have been demonstrated. We hope that this compact network that comprehensively considers the different properties of the objects in remote sensing images will become the new object detection baseline framework.

There are still room for improving the proposed method, for example, the contextual relationship between different objects in a remote sensing image can be considered in future work.

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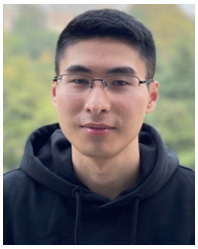
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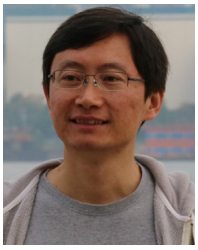
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